PROJECT AND TEAM INFORMATION

Project Title

IntelliPaper: Intelligent Question Paper Generation System

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

The conventional process of question-paper preparation is largely manual. Faculty members submit questions independently, after which the HOD or authorized paper setter has to search, verify, combine, and organize questions to create an examination paper. As the number of questions and faculty members increases, this process becomes time-consuming and difficult to manage efficiently.
Manual preparation can also result in repeated questions, poor unit coverage, unbalanced difficulty levels, and accidental reuse of recently used questions. Maintaining the history of previously generated papers and tracking how frequently individual questions have been used is particularly difficult without a centralized system.
The proposed IntelliPaper: Intelligent Question Paper Generation System aims to address these problems by providing a centralized question repository and an automated question-paper generation mechanism. Faculty members can submit questions along with metadata such as subject, unit, topic, marks, difficulty, and Bloom's level. The HOD or paper setter can then specify the requirements of a question paper, after which the system filters eligible questions and prioritizes them according to their usage history.
The project is particularly motivated by the integration of Database Management Systems and Operating Systems concepts. DBMS concepts will provide reliable storage, querying, transactions, and data integrity, while OS concepts such as concurrency, synchronization, critical sections, locking, and scheduling will help handle simultaneous users and operations safely.
The overall goal is to reduce manual effort while producing balanced, reliable, and less repetitive question papers.
The underlying problem and motivation are consistent with the supplied project report.

State of the Art / Current solution (1 pt)

Currently, question-paper preparation is commonly performed using manually maintained question banks, documents, spreadsheets, or separate submissions from faculty members. Faculty members prepare and submit questions, while the HOD or paper setter manually selects suitable questions based on subject coverage, marks, difficulty, and previous experience.
Although digital question repositories can make storing and searching questions easier, the selection and final paper-generation process may still require significant manual effort. In particular, maintaining a systematic history of previously used questions and ensuring that recently or frequently used questions receive lower priority can be difficult.
The proposed system improves upon this approach by centralizing questions and their metadata in a relational database. Instead of manually searching through questions, the HOD enters paper requirements such as the number of questions, marks, units, and difficulty distribution. The system then filters eligible questions and ranks them using usage history.
A starting priority mechanism is:
Priority = 1 / (Times Used + 1)
Thus, questions that have never been used receive greater priority than questions that have been used frequently. The system can additionally consider last-used date, unit coverage, difficulty balance, and controlled randomization in future versions.

Project Goals and Milestones (2 pts)

0.1 Project Goals

The main goal of the project is to develop an intelligent system capable of generating question papers automatically from a centralized question repository while maintaining appropriate question diversity and data integrity. The specific goals are:

1. *Develop a centralized repository for questions submitted by multiple faculty members.*
2. *Store question metadata including subject, unit, topic, marks, difficulty, and Bloom's level.*
3. *Provide role-based functionality for faculty, HOD/paper setter, and administrator.*
4. *Allow the HOD to specify question-paper requirements.*
5. *Automatically filter questions according to the specified requirements.*
6. *Implement a priority-based question-selection algorithm using question usage history.*
7. *Reduce repeated use of recently or frequently used questions.*
8. *Maintain generated-paper and question-usage history.*
9. *Apply DBMS concepts such as normalization, SQL, indexing, transactions, and constraints.*
10. *Demonstrate OS concepts such as concurrency, synchronization, critical sections, locking, and scheduling.*

0.2 Initial Milestones

0.2.1 Milestone 1 — Requirement Analysis

- *Study the existing manual question-paper process.*
- *Identify users and system requirements.*
- *Define functional and non-functional requirements.*

0.2.2 Milestone 2 — Database Design

- *Identify entities and relationships.*
- *Prepare the ER model.*
- *Design normalized relational tables.*
- *Define primary keys, foreign keys, and constraints.*

0.2.3 Milestone 3 — System Architecture

- *Design frontend, backend, database, and algorithm components.*
- *Define communication between the modules.*

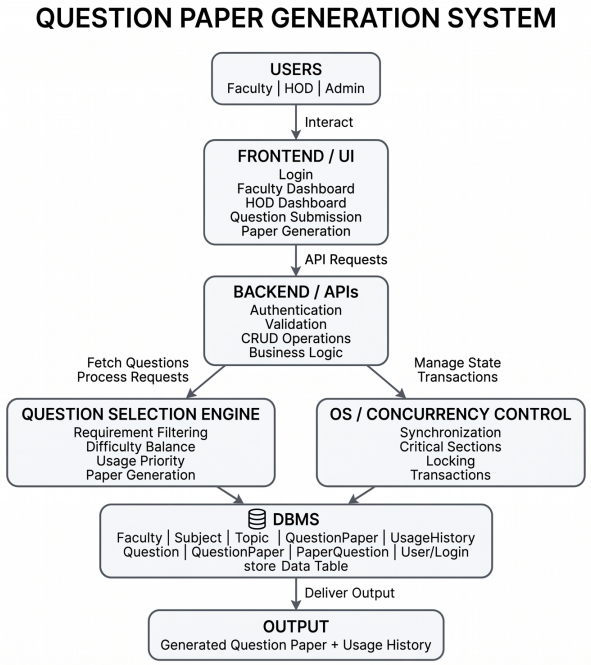
0.2.4 Milestone 4 — Core Algorithm

- *Implement question filtering.*
- *Implement usage-based priority calculation.*
- *Design paper-generation logic.*

Project Approach (3 pts)

The proposed system will follow a modular ***frontend–backend–database architecture****.*
The ***frontend*** *will provide role-based interfaces for faculty and HOD/paper setters. Faculty members will be able to log in and submit questions with relevant metadata. The HOD will be able to define paper requirements, search the question bank, generate a paper, review it, and regenerate or finalize it.*
The ***backend*** *will provide APIs for authentication, question management, database operations, filtering, and paper generation. It will validate incoming data and communicate between the frontend and database. CRUD operations will be used for managing faculty, questions, and related records.*
The ***DBMS layer*** *will use a relational database to store faculty, subjects, topics, questions, generated papers, paper-question mappings, usage history, and user information. The design will apply primary and foreign keys, normalization, constraints, SQL queries, joins, indexing, and transactions.*
The ***intelligent selection module*** *will first filter questions based on subject, unit, marks, difficulty, and approval status. Eligible questions will then be ranked according to usage history using the initial formula:*
Priority = 1 / (Times Used + 1)
The ***Operating Systems component*** *will address concurrent access when multiple faculty members submit or modify questions or when multiple paper-generation requests occur. Critical sections, synchronization, locking, transactions, race-condition prevention, and deadlock awareness will be considered.*
The project will be developed incrementally through requirement analysis, system design, implementation, testing, and evaluation. The team will use appropriate programming languages, a relational DBMS, frontend technologies, development tools, and Git/GitHub according to the final implementation choices.
The proposed workflow and OS/DBMS integration are supported by the project documentation.

System Architecture (High Level Diagram)(2 pts)



0.2.5 Architecture Explanation

The system consists of four major layers: ***Frontend, Backend, Intelligent Selection Engine, and Database****. Users interact with the frontend according to their roles. The frontend communicates with backend APIs, which validate requests and execute the required operations. The question-selection engine filters and ranks eligible questions before generating the paper. The relational database stores questions, metadata, generated papers, and usage history. Concurrency and transaction mechanisms ensure that simultaneous operations do not result in inconsistent data.*
The proposed database tables and their relationships are specified in the project report.

Project Outcome / Deliverables (1 pts)

The expected outcome of the project is a functional ***Intelligent Question Paper Generation System*** *that automates major parts of the question-paper preparation process.*
The primary deliverables will include:

1. ***Centralized Question Repository****for storing questions submitted by faculty.*
2. ***Relational Database****containing faculty, subjects, topics, questions, papers, and usage history.*
3. ***Faculty Module****for submitting, editing, and viewing permitted questions.*
4. ***HOD/Paper Setter Module****for defining examination requirements and generating papers.*
5. ***Question Filtering Module****for selecting questions according to subject, unit, marks, difficulty, and status.*
6. ***Intelligent Priority Algorithm****for reducing repeated selection of frequently or recently used questions.*
7. ***Question Paper Generation Module****for producing the final paper.*
8. ***Usage History Module****for maintaining records of previously used questions.*
9. ***Concurrency and Transaction Handling****to maintain data integrity during simultaneous operations.*
10. ***Testing and Evaluation Report****containing test cases and system results.*

The final system should demonstrate practical integration of DBMS and Operating Systems concepts while reducing manual effort and improving the reliability and diversity of generated question papers.

Assumptions

The system assumes that faculty members have authorized accounts and submit valid questions with complete metadata. The HOD or authorized paper setter is assumed to provide valid paper requirements, including subject, units, marks, number of questions, and difficulty distribution. Questions stored in the database are assumed to be reviewed or approved before being considered for paper generation. The system assumes a relational DBMS is available and that users have appropriate access permissions. Concurrent access by multiple users is expected, and database transactions and locking mechanisms are assumed to be available for maintaining data consistency.

References

1. *Silberschatz, A., Galvin, P. B., & Gagne, G., Operating System Concepts.*
2. *Korth, H. F., Silberschatz, A., & Sudarshan, S., Database System Concepts.*
3. *MySQL Documentation — Official documentation for relational database implementation.*
4. *PostgreSQL Documentation — Official documentation, if PostgreSQL is selected.*
5. *Documentation for the selected frontend/backend technologies.*
6. *References related to* ***Bloom's Taxonomy*** *for question classification.*
7. *Research papers related to automated question-paper generation and question-selection systems.*